#1 等价无穷小易错点

错误解法如下，为什么错？

【例 1】$\lim\limits_{x\to 0}\dfrac{\sin(x^2\sin\frac{1}{x})}{x}=\lim\limits_{x\to 0}\dfrac{x^2\sin\frac{1}{x}}{x}=0$

【例 2】$\lim\limits_{x\to 0}\dfrac{\tan x-x}{x^3}=\lim\limits_{x\to 0}\dfrac{x-x}{x^3}=0$

【例 3】$\lim\limits_{x\to 0}\dfrac{x\cos x-\sin x}{x^3}=\lim\limits_{x\to 0}\dfrac{x-\sin x}{x^3}=\lim\limits_{x\to 0}\dfrac{\frac{1}{6}x^3}{x^3}=\dfrac{1}{6}$

【例 4】$\lim\limits_{x\to 0}\dfrac{x\cos x-\sin x}{x^3}=\lim\limits_{x\to 0}\dfrac{x\cos x-x}{x^3}=\lim\limits_{x\to 0}\dfrac{\cos x-1}{x^2}=-\dfrac{1}{2}$

【例 5】$\lim\limits_{x\to 0}\dfrac{\ln(\sin^2 x+\mathrm{e}^x)-x}{x^2}=\lim\limits_{x\to 0}\dfrac{\sin^2 x+\mathrm{e}^x-1-x}{x^2}=1+\lim\limits_{x\to 0}\dfrac{\mathrm{e}^x-1-x}{x^2}=\dfrac{3}{2}$

没咋了26考研交流Q群：747259118　　没咋了26考研微信交流群请加V：meizhale888

正确解法如下：

【例 1】$\lim\limits_{x\to 0}\dfrac{\sin\left(x^2\sin\dfrac{1}{x}\right)}{x}=0$，原因如下：由于$\left|\sin\left(x^2\sin\dfrac{1}{x}\right)\right|\le\left|x^2\sin\dfrac{1}{x}\right|\le x^2$，所以$0\le\left|\dfrac{\sin\left(x^2\sin\dfrac{1}{x}\right)}{x}\right|\le|x|$，由夹逼准则知$\lim\limits_{x\to 0}\left|\dfrac{\sin\left(x^2\sin\dfrac{1}{x}\right)}{x}\right|=0$，故$\lim\limits_{x\to 0}\dfrac{\sin\left(x^2\sin\dfrac{1}{x}\right)}{x}=0$.

【例 2】$\lim\limits_{x\to 0}\dfrac{\tan x-x}{x^3}=\lim\limits_{x\to 0}\dfrac{\frac{1}{3}x^3}{x^3}=\dfrac{1}{3}$

【例 3】$\lim\limits_{x\to 0}\dfrac{x\cos x-\sin x}{x^3}=\lim\limits_{x\to 0}\dfrac{x\cos x-x+x-\sin x}{x^3}=\lim\limits_{x\to 0}\dfrac{x(\cos x-1)}{x^3}+\lim\limits_{x\to 0}\dfrac{x-\sin x}{x^3}=-\dfrac{1}{3}$

【例 4】同上。

【例 5】$\lim\limits_{x\to 0}\dfrac{\ln(\sin^2 x+\mathrm{e}^x)-x}{x^2}=\lim\limits_{x\to 0}\dfrac{\ln(\sin^2 x+\mathrm{e}^x)-\ln\mathrm{e}^x}{x^2}=\lim\limits_{x\to 0}\dfrac{\ln(\sin^2 x+\mathrm{e}^x)-\ln\mathrm{e}^x}{x^2}=\lim\limits_{x\to 0}\dfrac{\ln(\frac{\sin^2 x}{\mathrm{e}^x}+1)}{x^2}=\lim\limits_{x\to 0}\dfrac{\frac{\sin^2 x}{\mathrm{e}^x}}{x^2}=1$

没咋了26考研交流Q群：747259118　　没咋了26考研微信交流群请加V：meizhale888